

## GATE XL

Life Sciences

### TREND-BASED MOCK PAPER

— 01 —

|                 |             |
|-----------------|-------------|
| Duration        | 3 Hours     |
| Total Questions | 65          |
| Maximum Marks   | 100         |
| Sections        | 10 subjects |

#### Difficulty Distribution

Easy: 26 questions (40%)

Moderate: 30 questions (46%)

Difficult: 9 questions (14%)

# GENERAL INSTRUCTIONS

GATE XL — Trend-Based Mock Paper

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## 1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

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## 2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
  - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
  - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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## 3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
  - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
  - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
  - NAT questions carry no negative marking.
  - MSQ questions carry no negative marking.
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## 4. How to Answer

- For MCQ: Select exactly one option.
  - For MSQ: Select one or more options. Partial credit is NOT awarded.
  - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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## 5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

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## 6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

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## 7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
  - The Question Booklet contains multiple pages. Verify the page count.
  - Immediately report any discrepancy in the Question Booklet to the invigilator.
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**Q1.** 1M [-0.33]

[Cell Biology]

Powerhouse of cell:

- (A) Nucleus
- (B) Mitochondria
- (C) Ribosome
- (D) ER

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**Q2.** 1M [no -ve]

[Evolution]

Natural selection proposed by:

\_\_\_\_\_  
*Enter your answer here*

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**Q3.** 1M [-0.33]

[Engineering Mathematics]

Mean of normal distribution is:

- (A) Median
- (B) Mode
- (C) Both
- (D) None

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**Q4.** 1M [-0.33]

[Microbiology]

Penicillin from:

- (A) Fungus
- (B) Bacteria
- (C) Virus
- (D) Algae

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**Q5.** 1M [-0.33]

[Evolution]

Natural selection proposed by:

- (A) Lamarck
- (B) Darwin
- (C) Mendel
- (D) Wallace

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**Q6.** 1M [-0.33]

[Cell Biology]

Powerhouse of cell:

- (A) Nucleus
  - (B) Mitochondria
  - (C) Ribosome
  - (D) ER
-

**Q7.**

1M [-0.33]

*[General Aptitude]*

Choose synonym of 'abundant':

- (A) Scarce
- (B) Plentiful
- (C) Minimal
- (D) Rare

**Q8.**

1M [no -ve]

*[Cell Biology]*

Powerhouse of cell:

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*Enter your answer here***Q9.**

1M [-0.33]

*[Cell Biology]*

Powerhouse of cell:

- (A) Nucleus
- (B) Mitochondria
- (C) Ribosome
- (D) ER

**Q10.**

1M [-0.33]

*[Botany]*

Photosynthesis produces:

- (A) O<sub>2</sub>
- (B) CO<sub>2</sub>
- (C) N<sub>2</sub>
- (D) H<sub>2</sub>

**Q11.**

1M [no -ve]

*[Botany]*

Photosynthesis produces:

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*Enter your answer here***Q12.**

1M [no -ve]

*[Microbiology]*

Penicillin from:

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*Enter your answer here***Q13.**

1M [-0.33]

*[Evolution]*

Natural selection proposed by:

- (A) Lamarck
- (B) Darwin
- (C) Mendel
- (D) Wallace

**Q14.**

1M [-0.33]

*[Ecology]*

Trophic level of producers:

- (A) 1st
- (B) 2nd
- (C) 3rd
- (D) 4th

**Q15.**

1M [-0.33]

*[Evolution]*

Natural selection proposed by:

- (A) Lamarck
- (B) Darwin
- (C) Mendel
- (D) Wallace

**Q16.**

1M [-0.33]

*[Evolution]*

Natural selection proposed by:

- (A) Lamarck
- (B) Darwin
- (C) Mendel
- (D) Wallace

**Q17.**

1M [-0.33]

*[Zoology]*

Heart of mammal is:

- (A) 2-chambered
- (B) 3-chambered
- (C) 4-chambered
- (D) 5-chambered

**Q18.**

1M [-0.33]

*[Ecology]*

Trophic level of producers:

- (A) 1st
- (B) 2nd
- (C) 3rd
- (D) 4th

**Q19.**

1M [no -ve]

*[Cell Biology]*

Powerhouse of cell:

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*Enter your answer here*

**Q20.**

1M [-0.33]

*[General Aptitude]*

Choose synonym of 'abundant':

- (A) Scarce
- (B) Plentiful
- (C) Minimal
- (D) Rare

**Q21.**

1M [-0.33]

*[Zoology]*

Heart of mammal is:

- (A) 2-chambered
- (B) 3-chambered
- (C) 4-chambered
- (D) 5-chambered

**Q22.**

1M [-0.33]

*[Evolution]*

Natural selection proposed by:

- (A) Lamarck
- (B) Darwin
- (C) Mendel
- (D) Wallace

**Q23.**

1M [no -ve]

*[Cell Biology]*

Powerhouse of cell:

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*Enter your answer here***Q24.**

1M [-0.33]

*[Microbiology]*

Penicillin from:

- (A) Fungus
- (B) Bacteria
- (C) Virus
- (D) Algae

**Q25.**

1M [-0.33]

*[Botany]*

Photosynthesis produces:

- (A) O<sub>2</sub>
- (B) CO<sub>2</sub>
- (C) N<sub>2</sub>
- (D) H<sub>2</sub>

**Q26.**

1M [no -ve]

*[Evolution]*

Natural selection proposed by:

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*Enter your answer here***Q27.**

1M [-0.33]

*[Ecology]*

Trophic level of producers:

- (A) 1st
- (B) 2nd
- (C) 3rd
- (D) 4th

**Q28.**

1M [-0.33]

*[Zoology]*

Heart of mammal is:

- (A) 2-chambered
- (B) 3-chambered
- (C) 4-chambered
- (D) 5-chambered

**Q29.**

1M [-0.33]

*[Microbiology]*

Penicillin from:

- (A) Fungus
- (B) Bacteria
- (C) Virus
- (D) Algae

**Q30.**

1M [no -ve]

*[Botany]*

Photosynthesis produces:

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*Enter your answer here***Q31.**

2M [no -ve]

*[Microbiology]*

Penicillin from:

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*Enter your answer here***Q32.**

2M [-0.66]

*[Microbiology]*

Penicillin from:

- (A) Fungus
- (B) Bacteria
- (C) Virus
- (D) Algae

**Q33.**

2M [-0.66]

*[Evolution]*

Natural selection proposed by:

- (A) Lamarck
- (B) Darwin
- (C) Mendel
- (D) Wallace

**Q34.**

2M [-0.66]

*[Evolution]*

Natural selection proposed by:

- (A) Lamarck
- (B) Darwin
- (C) Mendel
- (D) Wallace

**Q35.**

2M [-0.66]

*[Biochemistry]*

Enzyme specificity is:

- (A) High
- (B) Low
- (C) Zero
- (D) Variable

**Q36.**

2M [-0.66]

*[Zoology]*

Heart of mammal is:

- (A) 2-chambered
- (B) 3-chambered
- (C) 4-chambered
- (D) 5-chambered

**Q37.**

2M [no -ve]

*[Ecology]*

Trophic level of producers:

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*Enter your answer here***Q38.**

2M [no -ve]

*[Evolution]*

Natural selection proposed by:

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*Enter your answer here***Q39.**

2M [-0.66]

*[General Aptitude]*

Choose synonym of 'abundant':

- (A) Scarce
- (B) Plentiful
- (C) Minimal
- (D) Rare

**Q40.**

2M [-0.66]

*[Biochemistry]*

Enzyme specificity is:

- (A) High
- (B) Low
- (C) Zero
- (D) Variable

**Q41.**

2M [-0.66]

*[Ecology]*

Trophic level of producers:

- (A) 1st
- (B) 2nd
- (C) 3rd
- (D) 4th

**Q42.**

2M [-0.66]

*[Cell Biology]*

Powerhouse of cell:

- (A) Nucleus
- (B) Mitochondria
- (C) Ribosome
- (D) ER

**Q43.**

2M [no -ve]

*[Microbiology]*

Penicillin from:

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*Enter your answer here***Q44.**

2M [no -ve]

*[General Aptitude]*

Choose synonym of 'abundant':

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*Enter your answer here***Q45.**

2M [no -ve]

*[General Aptitude]*

Choose synonym of 'abundant':

- (A) Scarce
- (B) Plentiful
- (C) Minimal
- (D) Rare

**Q46.**

2M [-0.66]

*[Microbiology]*

Penicillin from:

- (A) Fungus
- (B) Bacteria
- (C) Virus
- (D) Algae

**Q47.**

2M [-0.66]

*[Zoology]*

Heart of mammal is:

- (A) 2-chambered
- (B) 3-chambered
- (C) 4-chambered
- (D) 5-chambered

**Q48.**

2M [-0.66]

*[Zoology]*

Heart of mammal is:

- (A) 2-chambered
- (B) 3-chambered
- (C) 4-chambered
- (D) 5-chambered

**Q49.**

2M [-0.66]

*[General Aptitude]*

Choose synonym of 'abundant':

- (A) Scarce
- (B) Plentiful
- (C) Minimal
- (D) Rare

**Q50.**

2M [no -ve]

*[Genetics]*

DNA replication is:

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*Enter your answer here***Q51.**

2M [-0.66]

*[Ecology]*

Trophic level of producers:

- (A) 1st
- (B) 2nd
- (C) 3rd
- (D) 4th

**Q52.**

2M [-0.66]

*[Microbiology]*

Penicillin from:

- (A) Fungus
- (B) Bacteria
- (C) Virus
- (D) Algae

**Q53.**

2M [-0.66]

*[Biochemistry]*

Enzyme specificity is:

- (A) High
- (B) Low
- (C) Zero
- (D) Variable

**Q54.**

2M [no -ve]

*[Zoology]*

Heart of mammal is:

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*Enter your answer here***Q55.**

2M [-0.66]

*[Microbiology]*

Penicillin from:

- (A) Fungus
- (B) Bacteria
- (C) Virus
- (D) Algae

**Q56.**

2M [-0.66]

*[Cell Biology]*

Powerhouse of cell:

- (A) Nucleus
- (B) Mitochondria
- (C) Ribosome
- (D) ER

**Q57.**

2M [no -ve]

*[General Aptitude]*

Choose synonym of 'abundant':

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*Enter your answer here***Q58.**

2M [-0.66]

*[Microbiology]*

Penicillin from:

- (A) Fungus
- (B) Bacteria
- (C) Virus
- (D) Algae

**Q59.**

2M [no -ve]

*[Microbiology]*

Penicillin from:

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*Enter your answer here*

**Q60.**

2M [no -ve]

*[Botany]*

Photosynthesis produces:

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*Enter your answer here***Q61.**

2M [-0.66]

*[Microbiology]*

Penicillin from:

- (A) Fungus
- (B) Bacteria
- (C) Virus
- (D) Algae

**Q62.**

2M [-0.66]

*[Botany]*

Photosynthesis produces:

- (A) O<sub>2</sub>
- (B) CO<sub>2</sub>
- (C) N<sub>2</sub>
- (D) H<sub>2</sub>

**Q63.**

2M [-0.66]

*[Zoology]*

Heart of mammal is:

- (A) 2-chambered
- (B) 3-chambered
- (C) 4-chambered
- (D) 5-chambered

**Q64.**

2M [-0.66]

*[Botany]*

Photosynthesis produces:

- (A) O<sub>2</sub>
- (B) CO<sub>2</sub>
- (C) N<sub>2</sub>
- (D) H<sub>2</sub>

**Q65.**

2M [-0.66]

*[General Aptitude]*

Choose synonym of 'abundant':

- (A) Scarce
- (B) Plentiful
- (C) Minimal
- (D) Rare